IC Imagine Public Charter School

8th Grade Mathematics · 2026–2027

Syllabus, Unit Outlines & Year Calendar

Contents

Welcome

Mr. Zachary Allen zacharyallen@icimagine.org

Welcome to 8th Grade Math! I am excited to work with you this year. Our class will be an engaging and challenging experience, designed to strengthen your skills and prepare you for high school mathematics. I believe in a focused approach to instruction, so that class time is used efficiently to practice and develop your mathematical abilities. I value a classroom that is comfortable and relaxed, yet structured and purposeful.

Together, we will explore a variety of topics, build a strong foundation, and develop real confidence along the way.

Class Schedule

Classes meet on an A/B block — about 90 minutes each — and typically run in three sections with two short breaks in between. Times and activities will vary.

General Schedule (A/B block — ~90 minutes)	General Outline (activities will vary)
1st Section (15–20 minutes)	Bell ringer · Collect assignments · Class check-in · Review previous class
2nd Section (30–35 minutes)	Direct instruction · Project-based learning · Presentations
3rd Section (30–35 minutes)	Group or individual practice · Related game/activity · Exit ticket / review

Grading

Grades in this course are standards-based: they describe how well a student has demonstrated mastery of each North Carolina Math 8 standard, and they are recorded in Infinite Campus. Assignments are grouped into categories and weighted by their purpose in the learning cycle.

Category	Weight	Purpose
Formatives	**0%**	Practice, feedback, and snapshots of the learning process.
Quizzes & Projects	**10%**	Intermediate check-points for applying skills.
Unit Assessments	**90%**	Final evidence of standard mastery within a unit.

How scores are calculated — the Decaying Average

Infinite Campus uses a decaying average rather than a simple mean, so a student's most recent work carries the most weight. As students learn and improve, newer evidence outweighs earlier attempts:

- Most recent score — **100% weight**

- Next most recent — **65%**
- Third most recent — **≈ 42% (65%²)**
- Fourth most recent — **≈ 27.5% (65%³)**

Cumulative grading

The grade book is cumulative and does not reset each quarter. Evidence of learning carries from Quarter 1 through Quarter 4, and a standard's proficiency level is continuously refined as new evidence is added. This means early-year struggles will not permanently lower a final grade once a student demonstrates mastery.

Final grade roll-up

Individual standards are scored with the decaying average across all four quarters. The standards within a domain are averaged to produce a domain score, and the final course score (on the 1–5 scale) is a weighted average of the domain scores, together with the EOG score where applicable.

Grading scale (1–5 mastery)

Each unit study guide ties its questions to these learning targets, so students always know what mastery looks like.

Grade	Description
5	**The student has shown mastery of the concepts, processes, and skills associated with the learning target and applied them to situations beyond what is expected for mastery.**
4	**The student has demonstrated their understanding of the concepts, processes, and skills associated with the learning target and applied them with limited errors.**
3	**The student has demonstrated a basic or incomplete understanding of the concepts, processes, and skills associated with the learning target.**
2	**With help, the student has demonstrated a partial success with the concepts, processes, and skills associated with the learning target.**
1	**With help, the student has demonstrated no success with the concepts, processes, and skills associated with the learning target.**
0	**The student did not make an attempt to complete the learning target.**

Re-testing & extra credit

Re-testing and extra credit are handled on an individual basis. Quizzes and formative work are primarily for feedback and practice and carry little or no weight in the grade.

Rules, Norms & Procedures

8th Grade Norms

- One person speaking at a time

- Phones off and away
- Keep hands, feet, and objects to yourself
- Be an engaged, attentive listener
- Respect the space, materials, and one another

Classroom Color System

A color signal at the front of the room shows the current mode of class. Match your behavior to the color.

Red — Instruction & Testing	Yellow — Practice & Group Work	Green — Break
• At your assigned seat • Raise your hand to speak • No snacks • Check in for the restroom	• Work at seats or designated areas • Move and talk as the task requires	• Move around and talk freely • Reset before the next section

The break timer restarts any time expectations are not being met.

Daily Procedures

Arrival:

- Line up in the designated hallway area and wait to enter.
- Collect any papers from the cart.
- Check the agenda slide, get out your materials, and copy the agenda into your planner.
- Be in your seat when the bell rings.

During class:

- Sharpen pencils during non-lecture time.
- One snack during a designated break; clean up after yourself.
- Sign out and back in for the restroom — one person at a time, not during the first or last five minutes.
- Ask for help when you need it.

End of class:

- With about three minutes left, check off your agenda and note what to finish.
- Pick up trash around you, clean your workspace, and push in your chair.
- Wait for dismissal, then follow the hallway transition plan.

Other Expectations

- Phones, smartwatches, and similar items not turned in will be held.
- Earbuds and headphones stay away unless you are given permission; music only with permission.
- Keep your laptop closed until you are asked to open it.
- Be quiet during announcements.

- Raise your hand to speak, and ask permission before getting materials.
- Stay in your seat until the bell; do not pack up until instructed.
- Be mindful of the posted procedures for the school nurse and counselors.

Behavior

Please follow the norms, color system, and expectations above. Otherwise, you will be held to the standards in our DIVE matrix. I use a discreet card system to redirect behavior: students are quietly reminded with a card, and after two cards a student receives a consequence. If you ever have questions or concerns, please let me know.

Attendance

Be proactive if you miss class. Email me and complete missing work promptly. Due dates for missing assignments or assessments are determined individually.

Materials & Technology

- Bring necessary materials to every class (notebook, pencils, calculator, etc.).
- Keep your laptop closed until instructed to open it.
- Music only with permission.

We use Infinite Campus for grading, attendance, and assignments, and IXL and DeltaMath for practice and enrichment. We will use the Desmos scientific and graphing calculators to prepare for the EOG; personal calculators are encouraged but not required. Please bookmark these links:

- DeltaMath — https://www.deltamath.com/
- IXL — https://www.ixl.com/dashboard
- Desmos — https://www.desmos.com/calculator

Student Support

If you need help, please advocate for yourself and ask. I am available in the morning and during lunch, and our science teacher will provide additional support this year as well. If you need anything further, let me know so I can help.

A Note to Families

I look forward to a great year. Please reach out any time with questions or concerns. Complete and return the section below by the end of next week.

Thank you,

Mr. Allen

Student Name: ____

Student Signature: ____

Parent/Guardian Name (printed): _________________________________

Parent/Guardian Signature: ____________________________________

Parent Cell: ___________________________ Parent Email: ___________________________

Unit Outlines — Year at a Glance

Dates follow the 8th Grade Math pacing calendar and are subject to change. Each unit ends with a standards-based unit assessment.

Unit 1 · The Real Number System (Aug 4 – Aug 26) · NC.8.NS.1–2

Review

- Operations with integers
- Positive, negative, absolute value
- Operations with fractions
- Converting numbers
- Multiplication facts
- Order of operations

Objectives

- Classify numbers within the Real Number System
- Convert fractions to decimals (terminating & repeating) and back
- Compare rational and irrational numbers
- Estimate non-perfect square roots on a number line and with an algorithm
- Sort and order a mixed set of numbers

Vocabulary

- Whole Numbers
- Integers
- Rational Numbers
- Irrational Numbers
- Square / Square Root
- Cube / Cube Root

Unit 2 · Exponents & Scientific Notation (Aug 27 – Sep 25) · NC.8.EE.1, 3, 4

Review

- Multiplication facts
- Square roots

Objectives

- Understand how exponents work
- Expand and condense numbers with exponents
- Apply the exponent rules (product, quotient, power-to-a-power)
- Evaluate negative and zero exponents
- Convert between standard and scientific notation
- Operate with numbers in scientific notation
- Recognize uses of scientific notation

Vocabulary

- Base
- Exponent
- Power
- Coefficient
- Monomial
- Scientific vs. Standard Notation
- Magnitude

Unit 3 · Linear Equations & Inequalities (Oct 6 – Nov 2) · NC.8.EE.7

Review

- Expressions vs. equations
- Evaluating expressions
- Number line and inequalities

Objectives

- Solve one- and two-step equations
- Solve multi-step equations
- Apply the distributive property
- Solve equations with variables on both sides
- Solve and graph inequalities on a number line
- Determine the number of solutions and check answers

Vocabulary

- Expressions / Equations
- Operations
- Combine Like Terms
- Distributive Property
- Reciprocal
- Inequality

Unit 4 · Angle Relationships (Nov 3 – Nov 24) · NC.8.G.5

Review

- How to read lines, points, and figures
- How to read the coordinate grid

Objectives

- Identify and solve problems involving angle relationships (including algebraic expressions)
- Identify and solve problems with angles formed when parallel lines are cut by a transversal
- Apply triangle angle-sum and exterior-angle relationships

Vocabulary

- Linear vs. Vertical Pair
- Complementary vs. Supplementary
- Parallel vs. Perpendicular
- Interior vs. Exterior
- Transversal

Unit 5 · The Pythagorean Theorem (Jan 5 – Jan 21) · NC.8.G.6–8

Review

- Identify different triangles
- Read triangles and their measurements

Objectives

- Identify and label the parts of a right triangle
- Solve for missing sides of a right triangle
- Use side lengths to determine whether a triangle is a right triangle (converse)
- Use the theorem to find distance on the coordinate plane
- Solve word problems with the Pythagorean Theorem

Vocabulary

- Right Angle / Right Triangle

- Leg vs. Hypotenuse
- Pythagorean Theorem
- Diagonal
- Free-Body Diagram
- Converse

Unit 6 · Functions & Linear Relationships (Jan 22 – Mar 2) · NC.8.F.1–4

Review
- Using the coordinate plane
- Proportional relationships

Objectives
- Define a function
- Identify and evaluate functions from tables, graphs, and mappings
- Determine domain and range
- Describe functions (increasing/decreasing, linear/non-linear)
- Find slope from graphs and from two points
- Write slope-intercept, point-slope, and standard form
- Solve systems of equations and word problems

Vocabulary
- Relation or Function
- Function Notation
- Input vs. Output
- Domain vs. Range
- Slope / Rate of Change
- Increasing vs. Decreasing

Unit 7 · Statistics (Mar 3 – Mar 19) · NC.8.SP.1–4

Review
- Mean, median, mode
- Reading the coordinate grid
- Slope-intercept form

Objectives
- Display and interpret bivariate data
- Create and complete two-way frequency tables
- Create and interpret scatter plots
- Identify trends and associations in scatter plots
- Determine the line of best fit and use it to make predictions

Vocabulary
- Bivariate Data
- Frequency Table
- Relative Frequency
- Scatter Plots
- Association
- Line of Best Fit

Unit 8 · Transformations & Dilations (Apr 6 – Apr 27) · NC.8.G.2–4

Review	Objectives
• Reading lines, points, and figures • Reading the coordinate grid • Ratios & proportions	• Identify congruent figures and corresponding parts • Translate, reflect, and rotate figures • Identify similar figures and find scale factor • Solve for missing sides of similar figures • Dilate figures on the coordinate plane

Vocabulary

- Congruence
- Translation
- Reflection
- Rotation
- Similarity
- Scale Factor
- Ratios and Proportions
- Dilations

Unit 9 · Volume: Cylinders, Cones & Spheres (Apr 28 – May 7) · NC.8.G.9

Review	Objectives
• Area and perimeter of basic shapes • Volume of basic shapes • How pi is determined	• Determine the volume of cylinders, cones, and spheres • Determine the volume of 3-D composite figures • Apply volume formulas to word problems

Vocabulary

- Perimeter vs. Area vs. Volume
- Cone
- Cylinder
- Sphere
- Pi

EOG Review (May 10 – May 28) · All NC.8 standards

Review	Objectives
• All units, spiraled • Standards-based study guides	• Spiral-review every domain (Number System, Expressions & Equations, Functions, Geometry, Statistics) • Practice full-length EOG-style problem sets • Target individual standards using DeltaMath and IXL data

Vocabulary

- Domain
- Spiral Review
- Cumulative Practice

EOG Testing (Jun 1 – Jun 11) · All NC.8 standards

Review	Objectives
• Testing strategies and pacing • Desmos calculator practice	• Administer the NC Math 8 End-of-Grade assessment • Demonstrate cumulative mastery of all NC.8 standards
Vocabulary • End-of-Grade (EOG) Assessment	

Year-at-a-Glance Calendar

8th Grade Mathematics — 2026–27 Year at a Glance

NC Math 8 · A/B Block · unit and break dates color-coded below

August 2026

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

September 2026

S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

October 2026

S	M	T	W	T	F	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

November 2026

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

December 2026

S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

January 2027

S	M	T	W	T	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

February 2027

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28						

March 2027

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

April 2027

S	M	T	W	T	F	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

May 2027

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

June 2027

S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

Legend

- 1 · Real Numbers
- 2 · Exponents & Sci. Notation
- 3 · Equations & Inequalities
- 4 · Angle Relationships
- 5 · Pythagorean Theorem
- 6 · Functions
- 7 · Statistics
- 8 · Transformations
- 9 · Volume
- EOG Review
- EOG Testing
- Break
- Imagination Day
- No School / Work Day

Dates follow the 8th Grade Math pacing calendar; subject to change. · IC Imagine · Mr. Allen